

Abundance Federation

Pilot 01: Two-Week Sprint Plan

Data Collection & Annotation Platform
Step-by-step implementation guide for Call 2 discussion

April 2026

Overview: The Two-Week Sprint

Rather than splitting the team across four areas that can't all run in parallel, we focus the first two weeks on the two workstreams that are fully self-contained and don't depend on domain experts or gold-standard data: Data Collection & Ingestion (Team A) and Annotation Platform (Team B).

QA scripting and analysis work are parked until we have real data flowing and domain expert input.

Team Structure

Team	Focus	Composition	Why This Pairing
Team A	Data Collection & Ingestion	1 × L3–4 + 1 × L1–2 volunteer	KoboToolbox setup is accessible at any level; the pipeline script needs stronger Python skills. The L1–2 person does the form design and field testing; the L3–4 person builds the ingestion script and coaches.
Team B	Annotation Platform	1 × L3–4 + 1 × L1–2 volunteer	Label Studio deployment needs Linux/Docker comfort; configuration and testing are accessible at any level. The L3–4 person handles deployment; the L1–2 person configures the project, creates accounts, and tests on mobile.

Two-Week Target

By the Week 4 call (when we expect academic advisors join), the team should be able to demonstrate:

- **A working data collection pipeline:** “We can photograph weeds, upload through KoboToolbox, and the images land in an organised folder structure with consistent naming.”
- **A working annotation platform:** “We can load images into Label Studio and annotate them through a browser on a phone.”
- **A test run:** “We collected X test images, put them through the full pipeline, annotated them in Label Studio, and here's what it looked like.”

This shows the professors: the human systems and tooling work. Which they can review based on their practical experience.

Time Commitment Summary

Role	Week 2	Week 3	Total	Notes
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	Hours	Hours		
Team A — L3–4	4–5 hrs	3–4 hrs	7–9 hrs	Heavier in Week 1 (pipeline script). Week 2 is testing and refinement.
Team A — L1–2	3–4 hrs	3–4 hrs	6–8 hrs	Steady across both weeks. Form design, field testing, documentation.
Team B — L3–4	5–6 hrs	2–3 hrs	7–9 hrs	Heavy in Week 1 (VM + deployment). Week 2 is lighter.
Team B — L1–2	2–3 hrs	3–4 hrs	5–7 hrs	Ramps up in Week 2 once the platform is running.
Coordinator	3–4 hrs	3–4 hrs	6–8 hrs	Drive setup, GitHub issues, meeting facilitation, domain expert outreach.

These are estimates, if things take longer that's ok but lets communicate with the team early. If life issues come up and you can't do something that's no problem but again please tell people then we can help.

Team A: Data Collection & Ingestion

Goal: Build a complete pipeline from “volunteer takes a photo on their phone” to “image lands in an organised Drive folder with correct naming and metadata.”

Tools: KoboToolbox (free hosted), Google Drive, Python (for the ingestion script)

Step-by-Step Plan

Step	Task	Detail	Who	Time	Prior Experience Needed
A1	Create KoboToolbox account	Go to kf.kobotoolbox.org , create a free account. Explore the interface. Familiarise yourself with how forms are built.	L1-2	30 min	None. Web browser only.
A2	Design the data collection form	Build the form in KoboToolbox with these fields: Volunteer ID (dropdown), Photo (image attachment, max 10MB), Farm location (dropdown of farm names — use “Test Farm” for now), Plot/area (optional text), Capture conditions (checkboxes: sunny, overcast, wet, dry, morning, afternoon), Notes (freeform text). GPS and timestamp are auto-captured. See Tooling Plan Section 3.1.1 for full field spec.	L1-2	1.5–2 hrs	None. KoboToolbox has a drag-and-drop form builder.
A3	Test the form on mobile	Download KoboCollect app (Android) or use the web form (iOS). Submit 5–10 test entries with real photos (any plants, not necessarily weeds). Check: Does GPS capture? Do images upload? Does the form work offline? Note any friction or confusion.	L1-2	1 hr	None. Just needs a smartphone.
A4	Review the form together	L1–2 walks L3–4 through the form. Discuss: Is anything missing? Are the field names clear? Would a non-technical volunteer understand this? Make revisions.	Both	30 min	None. This is a conversation.
A5	Export test data from KoboToolbox	Do a manual export: download the CSV + media bundle from KoboToolbox. Examine the structure: What does the CSV look like? How are images	L3-4 (L1-2 observes)	30 min	Basic comfort with CSV files and file management.

		named? Where is the GPS data? This informs the ingestion script design.			
A6	Write the ingestion pipeline script	Create scripts/ingestion/ingest.py. The script should: (1) Connect to KoboToolbox REST API and download new submissions, (2) Rename images using convention {farm}_{volunteerID}_{date}_{sequence}.jpg, (3) Append metadata to a master CSV tracking file, (4) Copy renamed images to the Google Drive Processed folder. Start simple — manual export + rename is fine for v0.1. API integration can come in v0.2. See Tooling Plan Section 3.1.3.	L3-4	3–4 hrs	Python (file I/O, CSV handling, os/pathlib). Familiarity with REST APIs is helpful but not essential for v0.1.
A7	Test the pipeline end-to-end	L1–2 submits 10–20 images through KoboToolbox. L3–4 runs the ingestion script. Verify: Do renamed images appear in Drive with correct naming? Is metadata captured in the tracking CSV? Fix any issues.	Both	1–1.5 hrs	L1–2: none (just submit photos). L3–4: ability to debug Python.
A8	Write the capture protocol (draft)	Create protocols/capture_protocol_v01.md. Document: How to install and use KoboCollect. How to photograph weeds (reference the Crowdsourcing doc: top-down shot, 45-degree shot, buffer zone, reference object, no digital zoom, no flash). Include example good/bad photos (can use the test images). Add a quick-reference card for printing.	L1-2 (L3-4 reviews)	1.5–2 hrs	None. This is writing, drawing from the Crowdsourcing Weed Data document.
A9	Field test with a second person	Give the capture protocol to someone who hasn't been involved (a friend, a housemate). Ask them to take 10 photos following only the written instructions. Note where they get confused or deviate. Update the protocol.	L1-2	1 hr	None. This is a usability test.
A10	Document the pipeline	Update the repo README with: how to run the ingestion script, where images end up, naming conventions, any known issues. Create a brief architecture note in docs/ showing the data flow: KoboToolbox → Export → Rename → Drive.	L3-4 (L1-2 reviews)	45 min	Basic markdown writing.

Team A: Week-by-Week Schedule

	Monday–Wednesday	Thursday–Friday	Weekend (optional)
Week 2	A1, A2, A3 (L1–2) A5, A6 begins (L3–4)	A4 (Both, 30 min call) A6 continues (L3–4) A7 first test (Both)	A6 finish, A7 re-test
Week 3	A8 (L1–2) A9 (L1–2) A10 (L3–4)	Final end-to-end test (Both) Protocol revisions Ready for academic demo	Buffer / catch-up

Team B: Annotation Platform

Goal: Get Label Studio running on a cloud server, accessible via any browser (including mobile), configured with a test project, and ready to receive images from Team A's pipeline.

Tools: Cloud VM (Hetzner CX22 or DigitalOcean, €5–15/month), Label Studio Community Edition, Caddy or Nginx (reverse proxy), Let's Encrypt (HTTPS)

Step-by-Step Plan

Step	Task	Detail	Who	Time	Prior Experience Needed
B1	Provision a cloud VM	Sign up for Hetzner (hetzner.com) or DigitalOcean. Create a small VM: 2 vCPU, 4 GB RAM, 50 GB SSD, Ubuntu 22.04 or 24.04 LTS. Note the IP address. Set up SSH access. Point a subdomain (e.g. label.abundancefederation.org) to the IP via DNS, or use a free subdomain service.	L3-4	30–45 min	Basic Linux command line. Comfort with SSH. Understanding of DNS (or willingness to follow a tutorial).
B2	Install Docker	SSH into the VM. Install Docker following the official Ubuntu guide (docs.docker.com). Verify with <code>docker --version</code> and <code>docker run hello-world</code> .	L3-4 (L1-2 follows along via screen-share)	20–30 min	Linux command line. Ability to follow documentation step-by-step.
B3	Run Label Studio	Run: <code>docker run -d -p 8080:8080 --name label-studio --restart unless-stopped -v label-studio-data:/label-studio/data heartexlabs/label-studio:latest</code> . Verify it's running: visit <code>http://[VM-IP]:8080</code> in a browser. Create an admin account.	L3-4	15–20 min	Docker basics (running containers). Can be done by following the command exactly.
B4	Set up HTTPS with Caddy	Install Caddy (caddyserver.com). Create a Caddyfile that reverse-proxies the subdomain to <code>localhost:8080</code> . Caddy handles HTTPS certificates automatically. Test: visit <code>https://label.yourdomain.org</code> and confirm Label Studio loads securely.	L3-4	30–45 min	Linux command line. Basic understanding of reverse proxies. Caddy is the simplest option — Nginx is an alternative if the L3–4 volunteer prefers it.
B5	Test mobile access	Open the Label Studio URL on an iPhone and an Android phone. Check: Does the login	L1-2	30 min	None. Just needs a phone and the URL.

		page load? Can you navigate the interface? Is the annotation view usable on a small screen? Note any issues.			
B6	Create a test project with placeholder labels	In Label Studio, create a new project called “Pilot 01 — Test.” Configure the labelling interface as single-label image classification. Use placeholder labels: Weed, Crop, Soil only, Unsure. (These will be replaced with the real taxonomy once the domain expert defines it.) Set the interface XML per Tooling Plan Section 5.1.2.	L1-2 (L3-4 guides)	30–45 min	None. Label Studio has a visual interface builder. The XML is short and documented.
B7	Import test images	Take 20–30 images from Team A’s test collection (or photograph some plants yourself). Upload them to the test project in Label Studio. Verify they display correctly.	L1-2	20–30 min	None. Drag-and-drop upload through the web interface.
B8	Create test user accounts	Create 4–6 user accounts in Label Studio (one per volunteer + a couple of spares for non-expert testers later). Use email/password authentication. Test logging in with a non-admin account and confirm the annotator can see the project and annotate images but not access admin settings.	L1-2	20–30 min	None. Done through the Label Studio admin panel.
B9	Test the annotation workflow	Log in as an annotator (not admin). Work through 10–20 images: select a label, submit, move to next. Time yourself. Is the flow intuitive? Are there confusing steps? Can you use it on mobile? Write up a short experience report: what works, what’s friction, suggested improvements.	L1-2	30–45 min	None. You are the test user.
B10	Test annotation export	As admin, export the annotations in JSON and CSV format. Examine the export: What fields are included? How are labels structured? Where are image references? This informs how the QA scripts will consume the data. Document the export format in docs/.	L3-4 (L1-2 observes)	30–45 min	Comfort reading JSON and CSV files. Understanding of data structures.
B11	Write deployment documentation	Document everything needed to reproduce the deployment: VM specs, Docker commands, Caddy configuration, domain setup, Label Studio version.	L3-4 (L1-2 reviews)	45 min–1 hr	Technical writing. Understanding of what was done in B1–B4.

		Include a “if the server goes down” recovery guide. Put this in scripts/setup/ and reference it from the README.			
B1 2	Cross-team integration test	Work with Team A to push images through the full pipeline: Team A submits photos via KoboToolbox → runs the ingestion script → images arrive in Drive → Team B imports them into Label Studio → annotates them. Verify the entire chain works end to end.	Both teams	1–1.5 hrs	Each person uses the tools they built. This is a system test, not a coding task.

Team B: Week-by-Week Schedule

	Monday–Wednesday	Thursday–Friday	Weekend (optional)
Week 2	B1, B2, B3, B4 (L3–4) B5 (L1–2, once B4 done)	B6, B7 (L1–2) B8 (L1–2) B9 (L1–2, start)	B9 finish, B10 (L3–4)
Week 3	B10 finish, B11 (L3–4) B9 revisions (L1–2)	B12: Cross-team integration test (All) Documentation polish Ready for academic demo	Buffer / catch-up

Summary: Steps by Skill Level Required

Use this to assign steps based on your volunteers' current level. Steps suitable for L1–2 produce real output while building familiarity with the tools — this is how people progress toward L3 and beyond.

The Data Science Volunteer Levels

These levels apply across the federation, not just this pilot. A new volunteer starts at L1 and progresses through demonstrated competence. This sprint's tasks are labelled by the minimum level needed.

Level	Title	Description	Typical Profile
L1	Observer / Beginner	Can follow written instructions, use web-based tools, and report what they see. Learning the domain and the processes.	New volunteer, any background. No technical prerequisites.
L2	Contributor	Can work independently on structured tasks, handle basic data files, write documentation, and test systems.	Has completed a few L1 tasks. Comfortable with spreadsheets and web tools.
L3	Practitioner	Can write scripts, deploy tools, design protocols, and debug problems. Coaches L1–2 volunteers.	Working data scientist, developer, or technically experienced volunteer (several projects at L2 level already done).
L4	Steward	Can design and run a pilot independently. Defines quality frameworks, mentors L3 volunteers, and liaises with domain experts.	Experienced data scientist who has completed at least one full pilot cycle. Or volunteer who has sufficient experience built up.
L5–6	Expert / Advisor	Domain experts and academic advisors. Define scientific standards, review methodology, and validate outputs.	Professors, senior researchers, domain specialists.

Steps Suitable for L1–2 (No Prior Technical Experience Needed)

Step	Task	What They'll Learn
A1	Create KoboToolbox account	How field data collection platforms work
A2	Design the data collection form	Form design, metadata thinking, user experience
A3	Test the form on mobile	Mobile field data collection, edge cases

A4	Review the form (pair discussion)	Collaborative review, protocol thinking
A8	Write the capture protocol	Technical writing, translating research into instructions
A9	Field test with a second person	Usability testing, observing non-expert behaviour
B5	Test mobile access	Quality assurance, cross-device testing
B7	Import test images	Data ingestion from a user perspective
B8	Create test user accounts	User management, access control concepts
B9	Test the annotation workflow	Annotation UX, writing experience reports

Steps Suitable for L2–3 (Basic Technical Comfort)

Step	Task	Skills Used
A5	Export and examine test data	CSV files, file management, data inspection
A10	Document the pipeline	Markdown writing, technical communication
B6	Create a test project with placeholder labels	Following configuration docs, basic XML (copy-paste level)
B10	Examine annotation export	Reading JSON/CSV, understanding data structures
B11	Write deployment documentation	Technical writing, system documentation

Steps Requiring L3–4 (Solid Technical Skills)

Step	Task	Skills Used
A6	Write the ingestion pipeline script	Python (file I/O, CSV, os/pathlib), REST APIs (optional for v0.1)
A7	Test and debug the pipeline	Python debugging, end-to-end system thinking
B1	Provision a cloud VM	Linux command line, SSH, DNS basics
B2	Install Docker	Linux package management, Docker fundamentals
B3	Run Label Studio in Docker	Docker container management
B4	Set up HTTPS with Caddy	Reverse proxy concepts, TLS certificates, Linux config files

Cross-Team Step (Everyone)

Step	Task	Skills Used
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B12	Cross-team integration test	Using the tools each team built. System-level thinking. No coding required.
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Coordinator Tasks — Parallel Track

While the two teams work on their areas, the coordination layer is also addressed:

When	Task	Time
Week 2	Finalise Google Drive folder structure (Raw_Uploads, Processed/{FarmName}, Gold_Standards, Exports, Protocols, Reports). Share with team.	30 min
Week 2	Create GitHub issues for all steps (A1–A10, B1–B12). Assign to team members based on role discussion at Call 2.	45 min
Week 2	Reach out to domain experts. Even if they can't engage fully yet, get: a rough target weed list (5–10 species), any existing reference images, their availability for Week 4–5.	1–2 hrs
Week 2	Confirm 3–4 non-expert test volunteers for Week 5–6.	30 min
Week 2	Mid-week check-in message in group chat. Ask: "How's everyone getting on? Any blockers?"	10 min
Week 2	Weekly sync (30 min, Tuesday)	30 min
Week 3	Weekly sync (30 min, Tuesday)	30 min
Week 3	Review Team A's capture protocol draft. Give feedback.	30 min
Week 3	Review Team B's deployment docs. Verify you could reproduce the setup from the documentation alone.	30 min
Week 3	Prepare for academic advisor call: draft a one-page briefing showing what's been built, what's working, what we need from them.	1 hr
Week 3	Invite academic advisor to Week 4 call. Send briefing doc in advance.	15 min

What “Parked” Means for QA and Analysis

Parking QA and analysis doesn't mean ignoring them. It means doing the right amount of prep without building things that can't be tested.

What to Do Now (1–2 Hours Total)

- **Gold tracker design doc:** The QA-inclined team member writes a half-page document describing: What inputs does the gold tracker need? (Label Studio JSON export + gold-standard CSV.) What outputs should it produce? (Per-volunteer accuracy, confusion matrix, routing rates.) What thresholds trigger retraining? This doc lives in docs/gold_tracker_design.md.
- **Analysis notebook outline:** The analysis-inclined team member writes a half-page outline of what each notebook should contain. Dataset Overview: class distribution bar chart, per-farm counts, resolution histogram. Quality Metrics: gold accuracy over time, agreement heatmap. ML Baseline: model choice, train/val/test split strategy, evaluation metrics. This lives in docs/analysis_plan.md.

What to Wait For

- **Domain expert taxonomy:** Needed before the gold tracker can define “correct” labels.
- **Gold-standard images:** Needed before the gold tracker can calculate accuracy.
- **Real annotations:** Needed before analysis notebooks can run on real data.
- **Label Studio export format:** Step B10 will document this, giving the QA and analysis volunteers the format spec they need to write working code.

Once these dependencies are resolved (Weeks 4–5), the QA and analysis volunteers will have design docs ready and a clear understanding of the data format. They'll be able to write working code much faster than if they'd started speculatively in Week 1.

Risks and Contingencies

Risk	Likelihood	Impact	Mitigation
VM deployment takes longer than expected (DNS issues, Docker problems)	Medium	Medium	Have the L3–4 volunteer attempt B1–B4 on day one. If stuck after 2 hours, switch to a managed platform (e.g. Railway, Render) or use Label Studio's Docker Compose recipe. Don't let infrastructure block the team.
KoboToolbox doesn't work on school/work networks	Low–Medium	Low	Fall back to the web form (no app needed). If completely blocked, use Google Forms + manual GPS entry as a temporary fallback.
L1–2 volunteers feel overwhelmed	Medium	High	The steps are sequenced so L1–2 volunteers start with zero-prerequisite tasks (A1, A2, B5, B7). Early wins build confidence. If someone is struggling, the paired structure means the L3–4 person can take over without the project stalling.
Team A and Team B fall out of sync	Low	Medium	The cross-team integration test (B12) is scheduled for late Week 3. If one team is behind, the other can help — e.g., Team B can submit test photos for Team A's pipeline, or Team A can help test Label Studio on mobile.
Domain experts aren't available for Week 4–5	Medium	Medium	The tooling work is independent of domain expertise. If experts are delayed, the team continues with placeholder labels and extends the internal test phase. The infrastructure gets more robust; nothing is wasted.